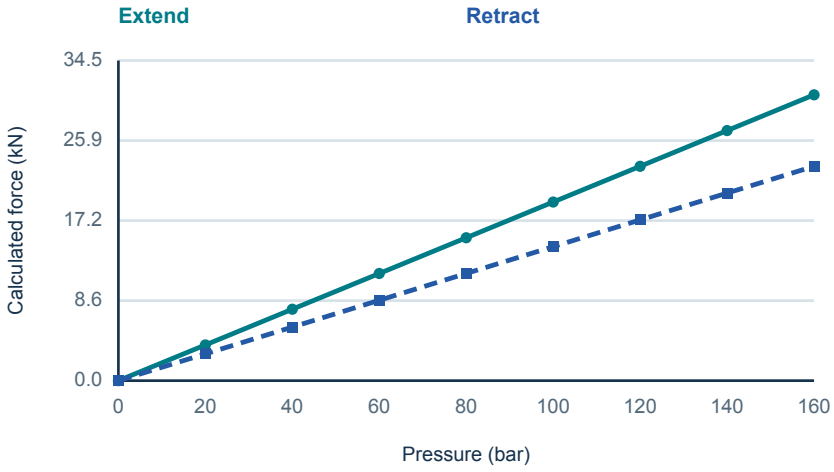
50 × 25 mm | Water cylinder

Bore 1.969 in (50 mm); rod 0.984 in (25 mm). Axial-load actuation.

Water pressure produces controlled linear motion. The rod reduces the pull area, so extension and retraction require separate force and speed checks. Select the stroke, mounting and control valves for the complete movement.

Calculated / published parameter	Value
Piston / annular area	19.635 / 14.726 cm²
Website maximum pressure	160 bar (2,321 psi)
Calculated push / pull at 160 bar	30.79 / 23.09 kN
Force model factor	0.98; calculation assumption, not tested system efficiency



bar	Push kN	Pull kN
0	0.00	0.00
20	3.85	2.89
40	7.70	5.77
60	11.55	8.66
80	15.39	11.55
100	19.24	14.43
120	23.09	17.32
140	26.94	20.20
160	30.79	23.09

Design & environmental advantages

Water operation removes hydraulic oil from the actuator circuit. The website identifies a stainless body and PEEK wear surfaces. Size and guide the mechanism to protect seals and rod alignment.

Efficiency & control

Choose bore for required force and flow for travel speed. Larger area increases force but needs more flow at the same speed. Load holding, cushioning and loss-of-supply behaviour depend on the selected valves and controls.

Engineering and purchasing information

State stroke, push/pull loads through travel, cycle rate, speed, mount style, rod end, orientation, external guidance, buckling length, water chemistry, temperature and any feedback/cushioning. The available range page supplies bore/rod and force data; overall dimensions, port sizes, stroke limits and selected seal rating require the configuration drawing.

Calculation: F = p × A × 0.98 / 100, with p in bar, A in cm² and F in kN; zero return-side gauge pressure. Return pressure, friction and dynamic loading reduce available force. 1 kN = 224.809 lbf; 1 bar = 14.5038 psi.

Calculate travel flow and water volume

At an illustrative speed of 0.10 m/s, calculated inlet flow is 11.78 L/min extending and 8.84 L/min retracting. Per 100 mm of stroke, swept volume is 0.196 L on extension and 0.147 L on retraction. These are geometric calculations, excluding leakage.

Q (L/min) = 6 × A (cm²) × speed (m/s). Swept volume (L) = A (cm²) × stroke (mm) / 10,000. The example speed is not a published speed rating.

WaterQuest cylinder range, accessed 17 Oct 2026. Forces independently recalculated from bore and rod geometry.